

Digital Transformation and Firm Productivity: A Managed Workflow Demonstration

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Abstract

This paper demonstrates a reproducible economics writing workflow using synthetic firm-level panel data. The empirical component estimates a difference-in-differences specification linking digital transformation status to firm log total factor productivity. The theoretical component uses Matlab to solve a simple AI adoption problem in which firms choose AI intensity by comparing productivity benefits with adoption costs. The data are entirely synthetic. The results should be read as a workflow demonstration, not as causal evidence about real firms or digital transformation policies.

Keywords: digital transformation; firm productivity; difference-in-differences; event study; AI adoption; synthetic data

1 Introduction

Digital transformation and artificial intelligence are changing how firms process information, organize production, and allocate managerial attention. A natural empirical question is whether firms that adopt digital technologies experience higher productivity after adoption. A related mechanism question is whether adoption depends on the balance between productivity benefits and implementation costs.

This paper uses a synthetic firm panel to demonstrate how an economics research workflow can connect Stata estimation, Matlab mechanism modeling, Jupyter notebooks, and formal L^AT_EX writing. Treated firms adopt digital technologies beginning in 2020, while control firms remain untreated. The baseline difference-in-differences estimate is approximately 0.1209, with a firm-clustered standard error of approximately 0.0047. These values are outputs of the synthetic data-generating process and must not be interpreted as real-world evidence.

2 Literature Review

The demonstration speaks to two strands of applied economics. The first studies information technology, automation, and artificial intelligence as productivity-changing technologies. ? show that IT-related productivity differences depend on organizational and managerial factors. ? frame automation as a force that both displaces and reinstates tasks, while ? interpret artificial intelligence through the economics of cheaper prediction. The Matlab mechanism in this paper follows that broad logic: firms adopt AI when the productivity value of adoption exceeds its costs.

The second strand concerns difference-in-differences and event-study research designs. ? emphasize inference problems in DID settings and motivate clustering standard errors at the relevant